

ANN10: MNIST Digit Recognition (CNN) – Detailed Code & Viva Explanation

Cell 1: Install & Import

Code:

```
try:
    import tensorflow as tf
except ImportError:
    !pip install tensorflow
    import tensorflow as tf
```

Explanation (what to say in viva):

This cell checks if TensorFlow is installed.

If not, it installs automatically.

This ensures code runs on any system.

Cell 2: Import Libraries

Code:

```
import numpy as np
import matplotlib.pyplot as plt
```

Explanation (what to say in viva):

Numpy is used for numerical operations.

Matplotlib is used for visualization.

Graphs help in understanding model performance.

Cell 3: Load Dataset

Code:

```
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()
```

Explanation (what to say in viva):

Loads MNIST dataset.

It contains handwritten digit images.

Used for training and testing.

Cell 4: Preprocessing

Code:

```
x_train = x_train/255.0
x_test = x_test/255.0
x_train = x_train.reshape(-1,28,28,1)
x_test = x_test.reshape(-1,28,28,1)
```

Explanation (what to say in viva):

Normalize pixel values between 0 and 1.

Reshape images to 28x28 with 1 channel.

Required for CNN input format.

Cell 5: Model (CNN)

Code:

```
model = tf.keras.Sequential([
    tf.keras.Input(shape=(28,28,1)),
    tf.keras.layers.Conv2D(32,(3,3),activation='relu'),
    tf.keras.layers.MaxPooling2D((2,2)),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(128,activation='relu'),
    tf.keras.layers.Dense(10,activation='softmax')
])
```

Explanation (what to say in viva):

Conv2D extracts features from image.

MaxPooling reduces image size.

Flatten converts 2D data to 1D.

Dense layers perform classification.

Softmax outputs probability for 10 classes.

Cell 6: Compile

Code:

```
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])
```

Explanation (what to say in viva):

Optimizer updates weights.

Loss function measures error.

Accuracy measures performance.

Cell 7: Train

Code:

```
history = model.fit(x_train, y_train, epochs=5)
```

Explanation (what to say in viva):

Model is trained for 5 epochs.

Each epoch improves learning.

Model learns patterns in digits.

Cell 8: Evaluate

Code:

```
model.evaluate(x_test, y_test)
```

Explanation (what to say in viva):

Model is tested on unseen data.

Checks how well model generalizes.

Cell 9: Prediction

Code:

```
prediction = model.predict(x_test[0].reshape(1,28,28,1))
```

Explanation (what to say in viva):

Predict digit for one sample.

Model outputs probabilities.

Cell 10: Output

Code:

```
print(np.argmax(prediction))
```

Explanation (what to say in viva):

Argmax gives index of highest probability.

This is predicted digit.